Pattern Recognition Assignment#4

May 10, 2023

Q1. Consider a three-layer feedforward neural network with two input neurons, two hidden neurons, and one output neuron. The architecture is shown below. The activation function of the hidden layer and the output layer is the Sigmoid function $f(x) = \frac{1}{1+e^{-x}}$. Given the initial model $\mathbf{w} = \left\{w_{11}^{(1)}, w_{12}^{(1)}, w_{21}^{(1)}, w_{22}^{(1)}, w_{11}^{(2)}, w_{12}^{(2)}\right\} = \{0.2, 0.3, 0.4, 0.5, 0.6, 0.7\}$, where $\left\{w_{11}^{(1)}, w_{12}^{(1)}, w_{21}^{(1)}, w_{22}^{(1)}\right\}$ correspond to the input-to-hidden layer weights and $\left\{w_{11}^{(2)}, w_{12}^{(2)}\right\}$ correspond to the hidden-to-output layer weights, one can yield the optimization using the stochastic backpropagation algorithm. Considering the training sample $\mathbf{x} = [2, 4]^\top$ with the target $t = 1$, what is the resulting model $\mathbf{w}$ after training the neural network with the sample (learning rate $\eta = 1$)? (Tips: you can round the results to the fourth decimal place).

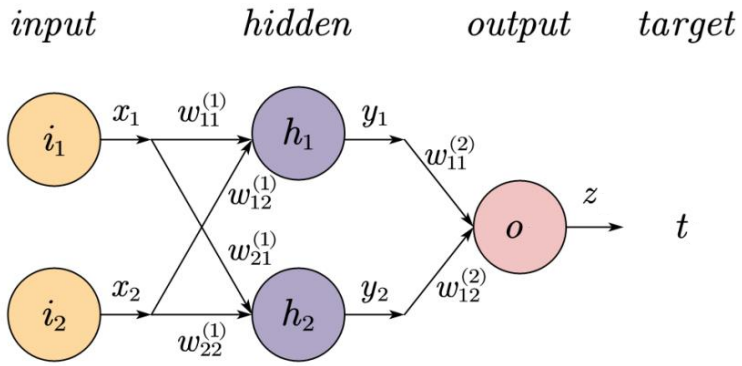


Figure 1: The architecture of a three-layer feedforward neural network.